

The nine-step reproducible screening pipeline (§2)

Quantum-chemical setup

1. Structures

SMILES → RDKit + MMFF
GFN2-xTB optimize
(drug & carrier)

2. Adsorption geometry

Place 3.2 Å above LOCAL
surface, 4 orientations,
optimize every complex

3. Interaction energies

$\Delta E_{\text{int,SP}}$ (frozen) &
 ΔE_{ads} (relaxed);
contact < 1.9 Å = chemisorb

Structure & interaction analysis

6. Descriptors

Conceptual-DFT indices
(η , S, χ , μ , ω)
from the same xtb output

5. Docking

AutoDock Vina vs. disease
target; pose-recovery
checked by redocking

4. $\Delta\rho$ map

GFN2-xTB densities
(Multiwfn) → isosurface
(ChimeraX)

Surrogate modelling & validation

7. Surrogate model

StandardScaler+RidgeCV
inside nested 5×5 CV;
1000× Y-scrambling

8. Applicability domain

OECD Principle 3;
Williams hat-matrix
leverage

9. Reproducibility

run_entire_<system>
_study.py; code + data
on GitHub & Zenodo